Bluetooth Home Automation using AT89S52 (8051)

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Abstract

This project presents a Bluetooth-based Home Automation System using the AT89S52 (8051) microcontroller, HC-05 Bluetooth module and a 2-channel relay module. A smartphone sends commands via Bluetooth, the microcontroller receives them through UART communication and controls a DC motor (fan) and an AC bulb.

Objectives

• Design a wireless home automation system.
• Interface HC-05 with AT89S52 using UART.
• Control electrical loads through relays.
• Develop firmware in Embedded C.

Hardware Components

AT89S52 Development Board, HC-05 Bluetooth Module, 2-Channel Relay Module, DC Motor, AC Bulb, USBasp Programmer, Breadboard, Jumper Wires, 5V Supply.

Software Tools

Keil uVision IDE, Embedded C, Serial Bluetooth Terminal (Android).

System Architecture

Smartphone → HC-05 → UART → AT89S52 → Relay Module → Fan/Bulb

Pin Connections

HC-05 TX → P3.0 (RXD)
HC-05 RX → P3.1 (TXD)
Relay IN1 → P2.0
Relay IN2 → P2.1
VCC → +5V, GND → GND

Bluetooth Commands

A = Bulb ON
a = Bulb OFF
B = Fan ON
b = Fan OFF

Working Principle

After Bluetooth pairing, the smartphone sends characters to the HC-05 module. The AT89S52 receives these characters via UART, interprets them, and drives the appropriate relay output. Since the relay module is active LOW, the firmware sets the corresponding output LOW to energize the relay and switch the appliance.

Applications

Smart homes, laboratory automation, educational demonstrations, wireless appliance control.

Advantages

Simple design, low cost, wireless operation, modular hardware, easy firmware modification.

Future Scope

Wi-Fi using ESP32/ESP8266, Android application, IoT dashboard, Google Assistant/Alexa integration, sensor-based automation.

Conclusion

The project successfully demonstrates Bluetooth-based control of home appliances using the 8051 microcontroller. It provides practical experience in Embedded C, UART communication, Bluetooth interfacing and relay control.